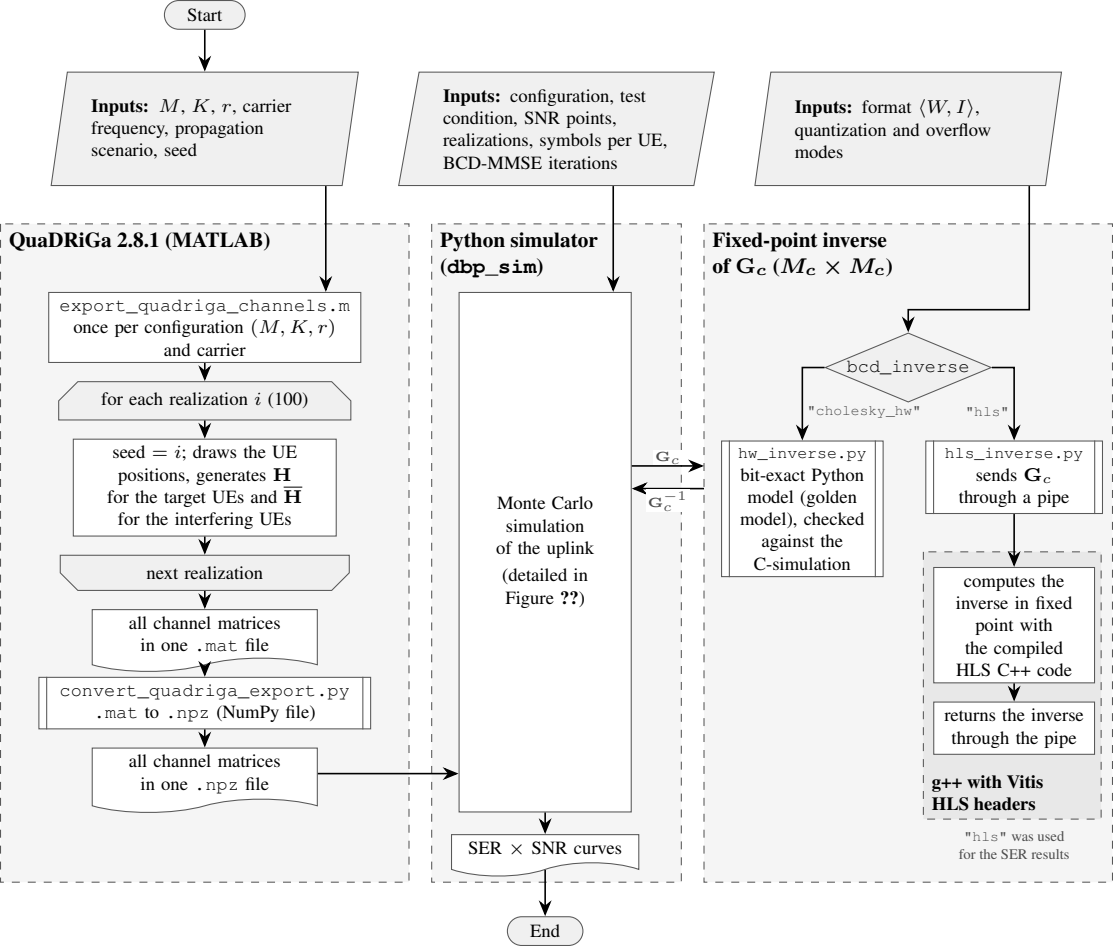

Start
Inputs: M, K, r, carrier frequency, propagation scenario, seed
Inputs: configuration, test condition, SNR points, realizations, symbols per UE, BCD-MMSE iterations
Inputs: format $\langle W, I \rangle$, quantization and overflow modes
QuaDRiGa 2.8.1 (MATLAB)
export_quadriga_channels.m once per configuration (M, K, r) and carrier
for each realization i (100)
seed $= i$; draws the UE positions, generates $\mathbf{H}$ for the target UEs and $\overline{\mathbf{H}}$ for the interfering UEs
next realization
all channel matrices in one .mat file
convert_quadriga_export.py .mat to .npz (NumPy file)
all channel matrices in one .npz file
Python simulator (dbp_sim)
Monte Carlo simulation of the uplink (detailed in Figure ??)
$\mathbf{G}_c$
$\mathbf{G}_c^{-1}$
SER × SNR curves
End
Fixed-point inverse of $\mathbf{G}_c$ ($M_c \times M_c$)
bcd_inverse
"cholesky_hw"
"hls"
hw_inverse.py bit-exact Python model (golden model), checked against the C-simulation
hls_inverse.py sends $\mathbf{G}_c$ through a pipe
computes the inverse in fixed point with the compiled HLS C++ code
returns the inverse through the pipe
g++ with Vitis HLS headers
"hls" was used for the SER results